

Planetary boundaries and data in HCI

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Abstract

This meet-up invites the Human-Computer Interaction community to figure out what sustainable data practices are in relation to planetary boundaries. The multiple ecological crises we are facing demand transformative change. We believe data is at the core of moving beyond talking about sustainability as an elusive concept, but what does this mean for data practices in HCI, and how do we move from talking about sustainability to consider planetary boundaries? The aim of this meet-up is to build a network, stimulate reflective discussions and practice-sharing among members of the HCI community on planetary boundaries and data practices in HCI, and envision actionable ways forward.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

Keywords

sustainability, planetary boundaries, data, data practice, doughnut model

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1 Motivation

The ongoing climate crisis underscores the urgent need for a fundamental shift in how we design and develop technologies, and how we co-exist with all living beings. Sustainable Human-Computer Interaction (SHCI) and Sustainable Interaction Design (SID) emerged around 15 years ago as a field that explicitly addresses these challenges, bringing design, technology, and sustainability concerns together [3, 4]. However, if we are truly committed to understand the unsustainable consequences of the technologies we design, we need to become better at working with different types of data (e.g., data on energy consumption and usage, climate data, behavioural data). These data serve as resources for understanding the multiple scales of environmental degradation and climate change, ranging from local experiences to global impacts.

The important role of data in the sustainability agenda is demonstrated in the following data practices: by making visible the environmental impacts of our practices [7, 14], illustrating the environmental footprint of emerging technologies [2, 8, 18], and expanding our understanding of the entangled, complex, and diverse perspectives that can be offered by data [9, 12, 20]. Using different types of data allows us to make sense of socio-ecological consequences across scales and contexts [13, 21, 23], underscoring an urgent need for the HCI community to develop data practices for engaging the multiple roles of data in enabling transformative change.

In this meet-up, we want to expand the discussion on the role of data for supporting pathways to stay within the social and planetary boundaries described by Raworth's Doughnut economics [19]. Traditionally, 'good' design has been equated with good usability

Activity	Time	Summary
Welcome and Framing	15 min	Introductions of organisers and attendees, introduce the Doughnut model and case, set expectations.
Sharing Examples	25 min	Groups share examples of data practices (based on a case or the participants' experiences) and map them in relation to planetary boundaries.
Network Building	20 min	Participants rotate to be inspired by each other's maps, meet new people, and integrate new perspectives.
Research Agenda Building	20 min	Plenary synthesis of group insights, clustering into broad themes
Wrap-Up and Commitments	10 min	Organisers summarise outcomes and invite continued collaboration.

and user experience, often framed through well-established design methodologies such as user-centred design [22]. We must expand these to account for broader considerations of the roles that data plays in cultivating socio-ecological change [11]. We need to understand long-term use, user behaviour, and collective practices, but also evaluate life cycle impacts, energy consumption, and supply chains. Last but not least, data are needed to evaluate the socio-economic consequences of design choices to ensure that transitions toward sustainability remain socially just. This requires an appreciation of, and capacity to work with, multiple complex types of data that reflect different perspectives and priorities.

The outcome of this meet-up is a discussion on how we can adopt more holistic framings of sustainability, such as the Doughnut economics model [10, 19], to enable a diversity of data types in the design of new technologies. In particular, the Doughnut model can be a useful tool for identifying a more focused research agenda beyond targeting climate change or addressing sustainability in broad strokes. In addition, it is a tool to remind researchers to be attentive to the complex interplay between ecological and social sustainability.

2 Meet-up themes and goals

This meet-up invites the HCI community to collectively explore how data can serve as a material for designing environmentally and socially sustainable technologies. We propose to discuss the following themes:

- (1) How might the Doughnut economics model be adopted as a framework for planning, positioning, and analysing projects that engage with data and sustainable technology design? In particular, how can it help move beyond broad notions of “sustainability” toward a more focused research agenda that attends to the interplay between ecological and social dimensions?
- (2) How can different data practices be explicitly considered throughout the design process to foreground the focus on sustainability?
- (3) How can we start reflecting on which particular type of data is being used and why? In particular, consider how traditional data on, e.g., usefulness and user experience, may fall short in addressing sustainability?
- (4) How can we adopt a stratified perspective, identifying different actors focusing on specific levels and areas of impact, e.g., grassroots and maker initiatives, academic communities focusing on higher education, designing and prototyping, awareness and providing target groups with tools for their

own sustainability assessments, and finally policy-level analyses leveraging large datasets?

Although we suggest the above topics for initiating the discussion, this meet-up aims to provide an open forum for the attendees to identify and discuss relevant topics of data practices in relation to planetary boundaries. Our primary aim in this meet-up is to provide an opportunity for people to build network, stimulate reflective discussions and practice-sharing among members of the HCI community on planetary boundaries and data practices in HCI. A further aim is to envision actionable ways forward.

3 Brief overview of activities

4 Meet-up community

We invite anyone in the HCI community interested in discussing future pathways for planetary boundaries and data in HCI, including the Sustainable and critical computing [6], Computing within limits [16], Climate action and HCI [24] communities, just to mention a few. Just like the organisers, who represent three different organisations and spanning wide in terms of interests within data practices in HCI, we invite participants from all areas of HCI. We expect to have both people who are already part of the community, as well as those who are starting to conduct research or practice in sustainable data practices in relation to planetary boundaries.

5 Meet-up Organisers

The organisers have previously collaborated as co-organisers in various related initiatives, such as a special interest group (SIG) at CHI'23 on more-than-human [25], workshops at DIS'24 and DIS'25 on challenges in decentering the Human [5, 17], and at the Aarhus Decennial Conference'25 on datafication of climate change [1, 15]. **Jonas Frich**, **Rikke Hagensby Jensen**, and **Eva Eriksson** are Associate Professors at the Department of Digital Design and Information Studies at Aarhus University, Denmark. They represent the Scandinavian Participatory Design tradition, sustainability, programming, more-than-human design, and data practices.

Lars Holmberg is Assistant Professor and **Elisabet M. Nilsson** is Associate Professor at the School of Arts and Communication (K3), Malmö University, Sweden. Together, they represent educational sciences, inclusive design, collaborative future-making, and AI.

Wolmet Barendregt and **Daisy Yoo** are Assistant Professors, and **Mathias Funk** is Associate Professor at the Department of Industrial Design at the Eindhoven University of Technology, The Netherlands. Together, they represent design for planetary boundaries, socio-ecological justice, data-driven design, and public welfare.

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